AI BASED PERSONAL FINANCE MANAGER

Conclusion, Future Scope and References

Department of Computer Science and Engineering, TKR College of Engineering & Technology

Batch C11 (CSE-C) · Academic year [ACADEMIC YEAR]

Table 1: Project team

S.No.	Student name	Roll number
1	Mekala Kalyan	23K91A05F0
2	Rajanala Sai Hemanth	23K91A05K3
3	Middala Surya Teja Reddy	23K91A05F2
4	Manga Akash	23K91A05E5

Project guide: N Padmavathi

Chapter and section numbers quoted in the text (for example Chapter 8 or Section 6.2) refer to the full project report, 17.Final Document. Figures and tables in this document are numbered from 1.

1. CONCLUSION

1.1 Summary

FINJARVIS is a working AI-based personal finance manager. A person signs up, records or imports transactions, and sees a dashboard with income, expenses, savings, net worth, a categorised spending breakdown, a next-month forecast, budget alerts and a 0-100 financial health score. An assistant answers questions from the person's own records, and reports can be downloaded as PDF, Excel or CSV. It is built from a React front end, a FastAPI back end with 33 routes, a service layer shared with the original Streamlit application and a SQLite database.

The work was verified by 38 pytest tests and a 49-check smoke test, and its machine-learning parts were measured on a probe set, a backtest, simulation and random profiles. The honest result is mixed: the platform and the explainable modules work, the categoriser and the forecast are simple and break in the ways reported, and the base paper's anomaly detector, budget recommender and deep models were not built.

1.2 Objectives against outcomes

Table 2: Objectives against outcomes

Objective	Outcome	Evidence
1 Secure accounts with transaction import and manual entry	Achieved	bcrypt, signed tokens, per-user scoping; manual entry and CSV/Excel import (916 of 1,000 rows imported, 84 bad rows skipped); 49 of 49 smoke checks
2 Categorise expenses automatically from descriptions	Partly achieved	71.7 percent on the probe set but 18.2 percent on unseen merchants; 51 seed examples; 11 categories
3 Forecast expenses and savings with time-series models	Partly achieved	Linear trend on monthly expenses only; no ARIMA or LSTM; MAPE 28.4 percent on the backtest, not better than a mean
4 Detect anomalous transactions	Not achieved	No anomaly detector exists; only the largest expense in a window is shown
5 Personalised budget recommendations and threshold alerts	Partly achieved	Alerts at 75, 90 and 100 percent and rule-based advice from the user's data; budgets are set by the user, no learned recommender
6 Interactive dashboard of spending, categories and forecasts	Achieved	13 pages in dark and light themes; dashboard 67 ms and analytics 72 ms median

1.3 Key findings

1. The categoriser is accurate where its vocabulary overlaps the seed list (100 percent on seed-like text, 97 percent on statement strings) and weak elsewhere (18.2 percent on unseen merchants), against 51.5 percent for the keyword table over the whole probe set.
2. Its mistakes are not random: 48 of 56 errors are labelled Transport (precision 0.273), because the model has no way to say it does not know.
3. A confidence threshold of 0.15 would keep 62.1 percent of the probe items at 94.3 percent accuracy, so abstaining is a practical improvement the app does not yet make.
4. One correction fixes a repeat merchant (18.2 to 100 percent) but forty-eight corrections move unseen-merchant accuracy only to 21.5 percent.
5. The linear-trend forecast is not better than a plain average for short histories: MAPE 28.4 against 21.4 percent on the demo backtest, and in simulation it overtakes the mean of all months at 7 months of history and the mean of the last three months only at 9 months, so the three-month minimum is too low.

6. The health score behaves sensibly (no decrease when the savings rate rises in 5,000 random profiles) but saturates: the savings-rate component stops changing at 20 percent, and its weights are not fitted.
7. The rule-based assistant is correct when it answers and silent about much of what users might ask: 14 of 20 answerable questions handled, 6 of 6 off-topic questions declined.
8. The application is fast for one user: every dashboard, list and analytics endpoint answers in under 82 ms median, reports in 136 to 209 ms, with login (305 ms) the slowest by design.

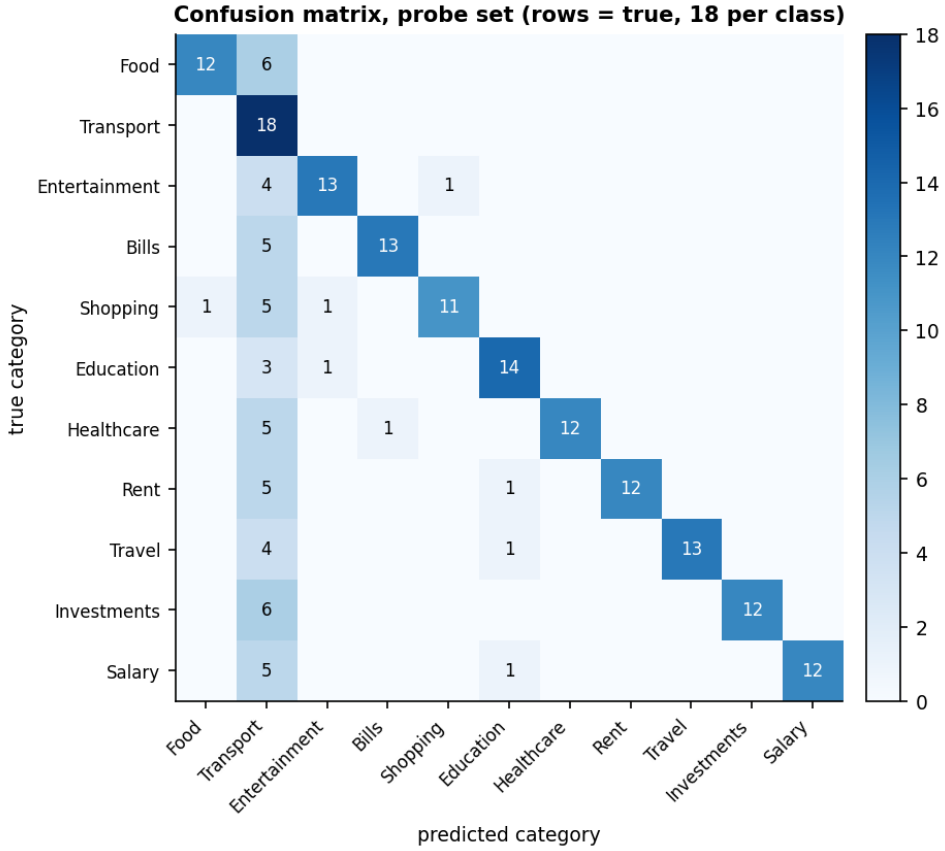


Figure 1: Confusion matrix of the shipped categoriser on the probe set: the Transport column collects the unseen-merchant errors.

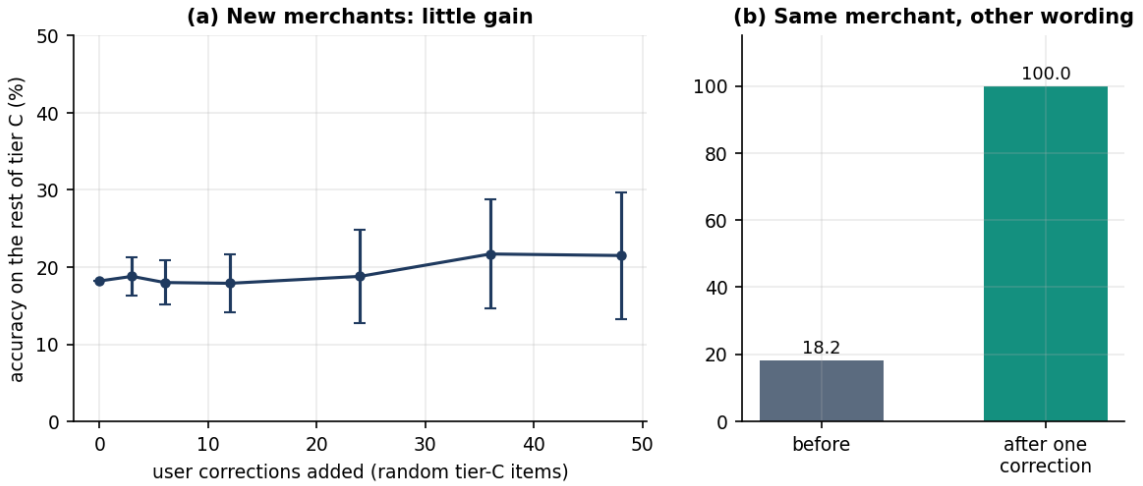


Figure 2: Learning from corrections: new merchants gain little, a repeat merchant is fixed by one correction.

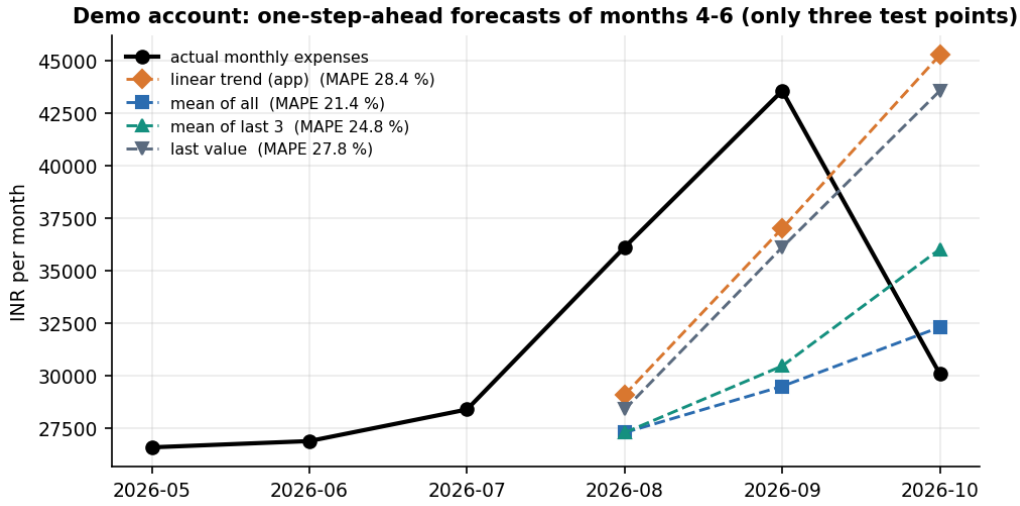


Figure 3: Forecasts for months 4-6 of the demo account with four methods; the actual spend fell in the last month.

1.4 Limitations

The evidence is a team-written probe set, fictional demo data and seeded simulation rather than field data, there was no user study, and the defects listed in the limitations section remain open. The results demonstrate a working, inspectable platform, not that its models are accurate on real transactions.

The measurements behind the key findings are shown below: how the categoriser behaves on the three probe tiers, how the forecast methods compare on the demo account, and how forecast error depends on the length of history in simulation. The probe set was written by the team and the simulation is seeded; neither is field data, and neither is a replication of the base paper's experiments.

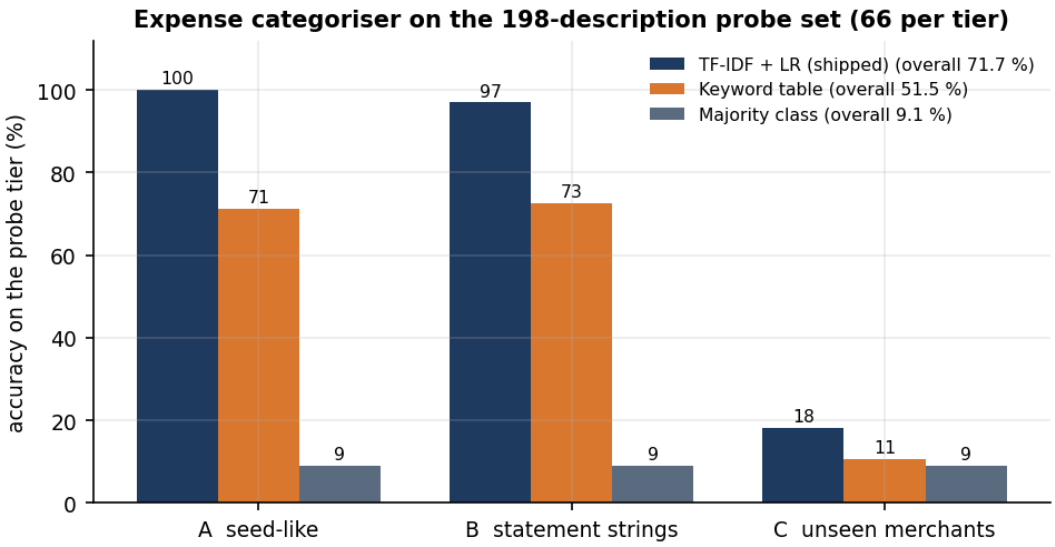


Figure 4: Accuracy of the shipped categoriser, the keyword table and the majority class on the three probe tiers.

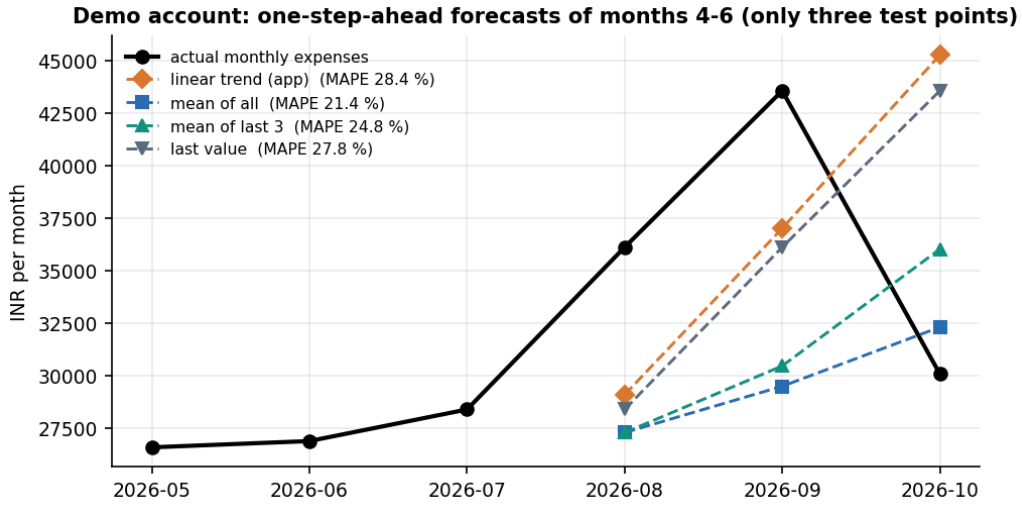


Figure 5: Demo account: forecasts of months 4-6 with four methods (three test points).

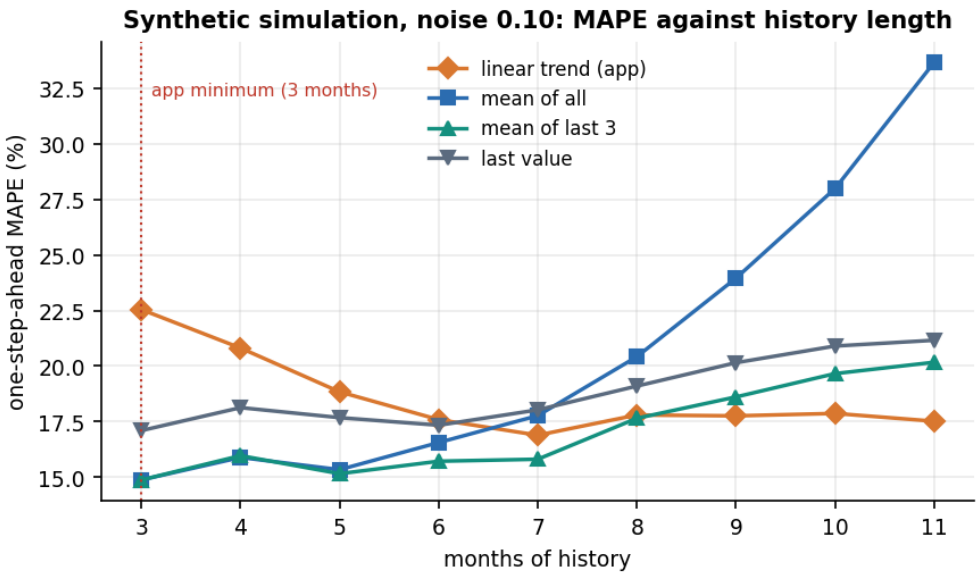


Figure 6: Synthetic simulation: forecast error against months of history (noise 0.10).

2. FUTURE SCOPE

2.1 Close the defects found

- **Let the categoriser abstain.** Return Other or ask the user when the top probability is below 0.15, which on the probe set keeps about 62 percent of items at about 94 percent accuracy; show the confidence next to the label.
- **Fall back to a mean for short histories.** Use the mean of the last three months until about seven months exist, and raise the minimum above three months.
- **Fix alert de-duplication and month ends** so that a budget crossing the same threshold in a later month raises a new alert, and compute month ends with the calendar.
- **Replace the token scheme** with a standard library, require `APP_SECRET_KEY` and refuse to start with the default, and add a login rate limit.
- **Batch the import commit** into one transaction so that a 1,000-row file takes seconds instead of tens of seconds.

2.2 Complete the objectives that were missed

- **Anomaly detection.** Start with a robust z-score per category, then an Isolation Forest on amount, category and day features, and measure precision and recall on injected outliers as the base paper does.
- **Savings forecasting.** Forecast income minus expenses, and compare a seasonal mean, exponential smoothing and ARIMA with the current line using a rolling-origin backtest.
- **A learned budget recommender.** Begin with a rule that proposes budgets from the last three months, and consider reinforcement learning only if real histories justify it.

2.3 Add features

- A **loan and debt tracker** with amortisation schedules and prepayment comparison, as named in the abstract.
- A **scheduler** for weekly or monthly digests and recurring-transaction detection.
- **Bank and SMS import** through statement parsers or an account-aggregator API, and merchant-name normalisation.

- A **larger labelled corpus** from consenting users and public sets, then a comparison of the current model with a linear model on character n-grams and a small transformer.
- **Testing the language-model layer** with a key, including answer faithfulness against the database.

2.4 Deploy and scale

- Move to PostgreSQL with Alembic migrations, run several workers behind a reverse proxy and keep secrets in the environment.
- Add load tests, automated browser tests and a dependency audit to the test suite, and monitor latency and error rates.

3. REFERENCES

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